

Beckett Dunlavy

(505)-738-9180 | beckett.dunlavy@gmail.com | [GitHub](#) | [LinkedIn](#)

EDUCATION

University of New Mexico

Albuquerque, NM

Major: Computer Science (Engineering) | Minor: German Language, GPA: 4.14/4.0

Aug. 2022 – Present

- Awards: Dean's List (5 semesters); Woodward Scholarship (4 years)
- Relevant Courses: High Performance Computing | Parallel Processing | Numerical Computing | Linear Algebra | Algorithms I/II | Design of Large Programs | Calculus I/II | Computer Architecture and Design | Software Engineering | Machine Learning | Operating Systems

Freie University

Berlin, Germany

European Studies Program, Intensive German Language Program (B2/C1)

Aug. 2024 – Dec. 2024

Google Cybersecurity Professional Certificate

Coursera Online

8-course cybersecurity certification program

June 2024

EXPERIENCE

Software Engineer Intern

May 2026 – Present

Tesla

Palo Alto, CA

- Engineered automation workflows to increase HPC cluster availability, reduce manual operational tasks, and improve system reliability.
- Built and maintained Kubernetes-based infrastructure tooling to support AI/ML cluster users, enabling more efficient deployment, monitoring, and resource management.

HPC Research Support

Jan. 2026 – May 2026

Center for Advanced Research Computing (UNM)

Albuquerque, NM

- Built, deployed, and troubleshooted scientific software on UNM high-performance computing clusters
- Investigated and resolved system-side issues across Linux systems and HPC environments

Research Assistant

May 2025 – May 2026

UNM Department of Computer Science, Advisor: [Dr. Amanda Bienz](#)

Albuquerque, NM

- Benchmarked the [Hypre](#) algebraic multigrid (AMG) linear solver on the [DeltaAI](#) HPC cluster at UIUC
- Extended existing C++ software to optimally leverage CPU and NVIDIA GH200 GPUs in AMG linear solvers

Faculty Assistant / Tutor

May 2024 – May 2025

UNM Department of Computer Science

Albuquerque, NM

- Created a curriculum to teach Git/GitLab to incoming and enrolled CS students
- Produced [educational YouTube videos](#) teaching the curriculum
- Tutored undergraduate computer science students in a variety of classes

PROJECTS AND ACTIVITIES

Student Cluster Competition | HPC, Linux, C++

Jul. 2025 – Nov. 2025

- Collaborated with teammates to optimize applications on VMs ran on the [Jetstream2](#) cloud system
- Built and ran climate applications across multiple nodes using slurm

UNM App Contest | Full stack development

Aug. 2025 – Oct. 2025

- Create an app to help voters understand their local elections and candidates

GitHub: [Scrabble](#) | [Word Search Solver](#) | [LCG cipher](#)

TECHNICAL SKILLS

Development: Java | C/C++ | Python | Jupyter Notebooks | MATLAB | Bash | Claude Code | Codex | Github Copilot

HPC: *Benchmarks* – HPL, STREAM | *Systems* – JetStream2, [CARC](#), DeltaAI | *Tools* – MPI, slurm, spack

Tools: Git/GitHub | LaTeX/Overleaf | Linux | MacOS | Windows | Microsoft Office Suite | Google Workspace